X-UAD — An Uncertainty-Aware and Self-Explaining Detector for Physical and Digital Face Attacks

Amrutha Babu
School of Computer Science and Engineering
Vellore Institute of Technology
Vellore, India
amrutha.babu2024@vitstudent.ac.in

Swethaa Premnath
School of Computer Science and Engineering
Vellore Institute of Technology
Vellore, India
swethaa.premnath2024@vitstudent.ac.in

Vividha Senthil Mohan
School of Computer Science and Engineering
Vellore Institute of Technology
Vellore, India
vividha.senthil2024@vitstudent.ac.in

Muthunagai S U*
School of Computer Science and Engineering
Vellore Institute of Technology
Vellore, India
muthunagai.su@vit.ac.in
*Corresponding author

Nivedha K
School of Computer Science and Engineering
Vellore Institute of Technology
Vellore, India
nivitha.k@vit.ac.in

Abstract

Face recognition systems can be fooled in two ways. Physical attacks show the camera a printed photo, a replayed video or a mask, while digital attacks change the captured image through face swapping, expression transfer, fully synthetic faces or adversarial noise. Most detectors handle only one of these groups. They also give a single score without saying how sure they are, where the fake evidence lies or why the decision was made. The aim of this work is a single detector that handles both groups and explains its own decisions. The proposed model, X-UAD, joins a central difference convolution stem and a frequency branch with a transformer that uses Adaptive Quantum Attention, which compares image patches through squared overlaps that suit repeating patterns such as screen stripes and printer dots. An evidential output layer gives an uncertainty score that is trained on blends of two attack types. A decoder marks the attack region and a rule-based module writes an explanation using only the measured values. On a simulated benchmark of eight attack types X-UAD reaches 0.42% ACER on seen attacks and 80.1% region-level IoU for localisation. It also gives the highest detection rate on unseen attack types at 77.5%. All tested methods still fail when the camera domain changes.

Index Terms

face anti-spoofing, presentation attack detection, deepfake detection, unified attack detection, vision transformer, quantum-inspired attention, evidential deep learning, uncertainty estimation, open-set recognition, explainable AI

I. Introduction

Face unlock and face verification systems can be attacked in front of the camera. A printed photo of the owner, a video replayed on a phone screen or a silicone mask can all be shown to the sensor. These are *physical* attacks. Each one leaves a physical trace behind, such as the dot pattern of a printer, the pixel grid of a screen that turns into a moiré pattern or the smooth waxy surface of silicone.

A face image can also be changed *after* a real photo has been taken. The identity can be swapped, the expression can be copied from another person, a face that never existed can be generated or a small pattern of noise can be added that flips the decision of a classifier. These are *digital* attacks. Their traces look very different, for example blending edges around a swapped face, mismatched resolution, checkerboard patterns from the upsampling layers of a generator and unusual peaks in the frequency spectrum.

Because the two kinds of traces look so different, the two research areas grew up separately. Face anti-spoofing focused on texture and depth cues [1]–[4], while face forgery detection focused on frequency and blending cues [5]–[8]. A real system needs one decision, one threshold and one record of why the decision was taken. Recent work [9]–[13] built joint datasets and joint detectors for both groups. On attack types seen during training these detectors reach an ACER of about 0.5 to 1.4%. When one attack family is held out during training the error rises to between 22 and 44% with large variation between runs [9]. Joint detection is therefore close to solved for known attacks but remains open for new ones.

Three more abilities are missing from most of this work. The first is a real sense of *not knowing*. A softmax output must spread all of its probability over the known classes, so a completely new attack is still given high confidence for whichever known class it looks most like. A second-order belief is needed that can stay close to empty when the input looks like nothing learned before. The second is *evidence localisation*. Depth-map supervision [1], [3] works for physical attacks because a photo or a screen is flatter than a real face. Depth says nothing about a swapped face that came from a live three-dimensional capture. The third is an *explanation tied to the decision*. Recent methods pass the face image to a multimodal language model and ask it to describe what looks fake [14], [15]. The language model never sees the internal state of the detector, so its text can mention artifacts such as a screen bezel that are not in the frame or were never used by the detector.

This paper presents X-UAD, a single model that detects physical and digital face attacks together, reports how uncertain it is, marks where the evidence lies and explains its decision with text built only from its own measured values. Three ideas make this possible. **Adaptive Quantum Attention (AQA)** adds a second scoring channel to normal attention. Much of the evidence for face attacks, such as a moiré beat, a checkerboard or a spectral peak, appears in the *squared size* of a projection and its sign carries no meaning. The new channel compares two tokens by the squared overlap of their unit vectors after a learnable rotation, which follows the Born rule of quantum measurement. A learned gate mixes this channel with the normal dot product. The rotations need only $\mathcal{O}(dL)$ angles instead of the $\mathcal{O}(d^2)$ weights of a linear layer. A **quantum evidence basis**, which is a set of orthonormal vectors built by the same kind of rotation, is measured against every spatial token. The resulting probabilities are used as classification features and are also shown unchanged as the explanation. **Pseudo-Unseen Evidence Suppression (PUES)** blends two training images from different attack families. No real camera or generator can produce such a blend, so it acts as a free stand-in for a new attack. The model is trained to give low evidence on these blends, which teaches the uncertainty output how an unknown input should look.

Every forward pass also writes a typed numeric **evidence record**. The text explanation is produced only from this record by a rule-based generator in which every sentence is guarded by a numeric condition, so the generator cannot invent facts. Since no capture-based dataset marks *where* the attack evidence lies, a difficulty-calibrated, identity-disjoint benchmark of eight attack families with pixel-level evidence masks was also built, together with seen-attack, leave-one-attack-out and cross-domain test protocols.

Section II reviews related work. Section III describes the proposed work including the benchmark and the model. Section IV reports the results and Section V concludes the paper.

II. Related Work

A. Physical Presentation Attack Detection

Early face anti-spoofing treated liveness as a texture classification problem. Auxiliary supervision [1] replaced the plain real or fake label with a pseudo-depth map and an rPPG signal as extra training targets. Central Difference Convolution [2] changed the first layers of the network so that they respond to fine texture that does not depend on lighting. It is still one of the strongest light backbones for physical attacks. Pixel-wise supervision was studied further in [3]. Later work tried to generalise across cameras and lighting through adversarial domain generalisation [16], single-side domain generalisation [17] and style-shuffled feature separation [4]. All of these methods assume that the attack happens in front of the camera. None of them can handle a real capture that was edited afterwards.

B. Digital Forgery Detection

FaceForensics++ [5] and Celeb-DF [6] are the main benchmarks of this area. The cue that transfers best is spectral. Generators leave periodic artifacts from their upsampling layers [18], which show up as peaks in the Fourier magnitude [7]. This is a large part of why simple detectors transfer well across generator types [8]. These detectors have no idea of a re-capture medium and fail on printed or replayed attacks.

C. Unified Physical and Digital Detection

UniAttackData [9] is an identity-consistent dataset in which the same person appears both as genuine and under every attack type, so identity cannot become a shortcut. It comes with a vision-language detector trained with teacher-student prompts. The CVPR 2024 and ICCV 2025 challenges built on it [10], [11] with nearly $2{,}900$ subjects and 54 attack types. Methods built on large pretrained multimodal backbones led the leaderboard while single-modality CNNs stayed above 20% ACER on the harder protocol. UniDefense [13] uses dual-space reconstruction for the same goal. None of these systems gives an uncertainty output, an evidence map or a real explanation.

D. Uncertainty and Open-Set Recognition

Maximum softmax probability [19] is a weak signal for unknown inputs because softmax is closed-set. Prior networks [20] and evidential deep learning [21], based on subjective logic [22], place a Dirichlet distribution over the class simplex so that total evidence and vacuity become outputs of the model. Outlier exposure [23] showed that even a rough stand-in for unknown

inputs during training helps more than any purely internal uncertainty score. Open-set single-domain generalisation was recently studied for anti-spoofing alone [24]. Evidential uncertainty has not yet been joined with unified physical and digital detection and no earlier method builds its outlier stand-in from the attack taxonomy itself.

E. *Explainability and Language-Model Forensics*

Attention rollout [25] and deletion and insertion tests [26] are the usual tools for checking whether an explanation reflects the decision of a model. FFAA [14] fine-tunes a multimodal LLM for open-world forgery analysis and M2F2-Det [15] joins CLIP with an LLM for interpretable forged-face detection. Both produce fluent text after the fact. The language model sees the image but not the internal evidence of the detector, so its account cannot be checked against the decision.

F. *Quantum-Inspired Attention*

Quantum vision transformers [27] and quantum self-attention [28], [29] replace linear projections with parameterised quantum circuits and report accuracy close to classical vision transformers with far fewer parameters. Quantum adaptive self-attention [30] places such a circuit inside one encoder layer and compares it with classical controls of equal size. This line of work is driven by parameter savings on generic image classification. It has not been applied to a task whose evidence is an interference pattern and it offers no explanation.

Each of these lines covers only part of what a deployed system needs, as summarised in Table I, where *partial* means that the ability exists only for one attack group or only as a proxy. The missing parts also depend on each other. Covering more attack families makes uncertainty more important. Uncertainty makes explanation more important, since a reviewer who is handed an uncertain sample needs to know what the model saw. An explanation is only worth trusting when it comes from the same internal evidence that produced the decision.

TABLE I
COMPARISON WITH PRIOR WORK

Representative work	Physical attacks	Digital attacks	Second-order uncertainty	Explicit unknown class	Evidence localisation	Text explanation	Explanation bound to the decision
CDCN++ [2], pixel-wise FAS [3]	yes	no	no	no	partial (depth proxy)	no	—
SSAN [4], SSDG [17]	yes	no	no	no	no	no	—
Frequency forensics [7], [8]	no	yes	no	no	no	no	—
UniAttackDetection [9], UniDefense [13]	yes	yes	no	no	no	no	—
Challenge winners [11]	yes	yes	no	no	no	no	—
Evidential and open-set [21], [24]	partial	no	yes	yes	no	no	—
FFAA [14], M2F2-Det [15]	no	yes	no	partial	partial	yes	no (post-hoc)
Quantum ViT [27], [29], [30]	no	no	no	no	no	no	—
X-UAD (proposed)	**yes**	**yes**	**yes**	**yes**	**yes**	**yes**	**yes**

III. PROPOSED WORK

A. *Overview*

X-UAD maps a face crop $\boldsymbol{x} \in \mathbb{R}^{3\times H\times W}$ to six outputs,

$$\boldsymbol{x} \longmapsto \big(\boldsymbol{\alpha},\ u,\ \boldsymbol{M},\ \boldsymbol{\Pi},\ r,\ \tau\big), \tag{1}$$

where in Eq. (1) $\boldsymbol{\alpha} \in \mathbb{R}^{K}_{>0}$ are Dirichlet concentrations over the $K = 8$ classes, $u \in [0, 1]$ is the vacuity, a single number that grows when the model has little evidence, $\boldsymbol{M}$ is the spoof-evidence map, $\boldsymbol{\Pi}$ is the quantum evidence spectrum over tokens and basis modes, r is the risk score and τ is the text explanation. All six come from one forward pass (Fig. 1). A central-difference stem and a spectral branch produce tokens, which are processed by a stack of AQA blocks. The quantum evidence basis is measured against the output tokens. An evidential head, a localisation decoder and a risk calibrator read out the results, after which the reasoning module turns the record into text. Table II lists the main symbols and abbreviations.

In Fig. 1 the stem supplies fine texture evidence that does not depend on lighting and the spectral branch supplies periodic evidence that cannot be seen at the token resolution. Each AQA block scores token pairs with a gated mix of the classical dot product and the Born-rule overlap between rotated unit vectors. The Born probabilities of the evidence basis feed the classifier *and* are exported unchanged as the explanation. This is what ties the explanation to the decision.

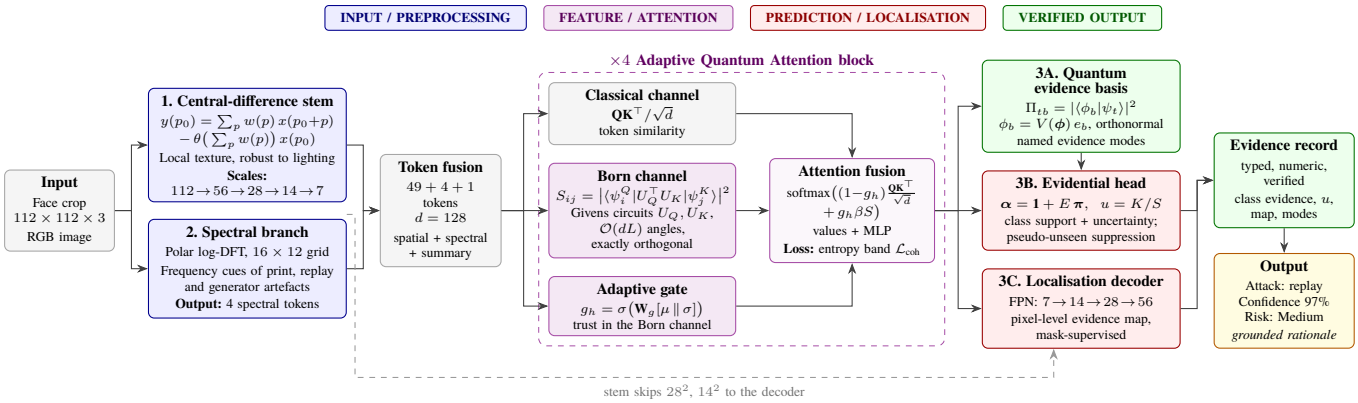


Fig. 1. Architecture of X-UAD

TABLE II
NOTATION AND ABBREVIATIONS

Symbol	Meaning
$\boldsymbol{x}$, K, s	Input face crop, number of classes, attack strength
$\boldsymbol{\alpha}$, u	Dirichlet concentrations, vacuity $u = K/S$
$\boldsymbol{M}$, $\boldsymbol{\Pi}$	Evidence map, quantum evidence spectrum
r, τ	Risk score, text explanation
d, L	Head dimension, depth of a rotation circuit
$G(\vartheta)$, U	Givens rotation, rotation circuit
S_{ij}	Born-rule overlap of tokens i and j
g_h, β	Gate of head h, Born-channel temperature
$\boldsymbol{A}$, $\mathcal{H}$	Attention matrix, its mean row entropy
$\lvert\phi_b\rangle$, B	Evidence basis vector, number of modes
$\boldsymbol{\pi}$, E, S	Class direction, total evidence, $S = K + E$
$\hat{p}_k$	Projected probability of class k
$\tilde{\boldsymbol{x}}$, ρ	PUES blend, PUES target ratio
ς_k, a	Severity of family k, evidence area
$\mathcal{L}$, w	Loss terms, loss weights

Abbreviation	Full form
X-UAD	Explainable Unified Attack Detection
AQA	Adaptive Quantum Attention
PUES	Pseudo-Unseen Evidence Suppression
CDC	Central Difference Convolution
APCER	Attack Presentation Classification Error Rate
BPCER	Bona fide Presentation Classification Error Rate
ACER	Average Classification Error Rate
AUC	Area Under the Curve
AUROC	Area Under the Receiver Operating Characteristic curve
EER	Equal Error Rate
ECE	Expected Calibration Error
IoU	Intersection over Union
PGD	Projected Gradient Descent
LFW	Labeled Faces in the Wild
MLLM	Multimodal Large Language Model

B. Benchmark Dataset

The capture-based datasets used for unified detection [9], [11] have restricted access. None of them marks *where* in the frame the attack evidence lies, because nobody knows exactly which pixels a real printer or generator changed. The benchmark used here therefore injects the evidence through known operators, so its support is known exactly. This gives pixel-level labels, full identity consistency and full reproducibility at the cost of external validity, since each operator is a model of a physical or generative process. To limit this risk every operator follows the physics or generator design it stands for, the benchmark is difficulty-calibrated and a second capture domain is kept for cross-domain testing. Configuration files that map the same taxonomy, protocols and code onto CASIA-FASD [31], OULU-NPU [32], SiW-Mv2 [33], CelebA-Spoof [34], FaceForensics++ [5], Celeb-DF-v2 [6] and UniAttackData [9] are released with the code.

Source faces. Domain A (LFW-Sim) is built from 12,794 identity-labelled, tightly cropped grayscale face photos of 5,616 subjects from the Labeled Faces in the Wild (LFW) database [35], obtained from its public distribution site [36]. Splits are assigned by a stable hash of the subject identity, so train, validation and test never share a person. Every source face gives exactly one genuine sample and one sample of each of the eight attack families. Identity therefore carries no information about the label. The grayscale source also removes colour shortcuts and forces every method to rely on texture, geometry and frequency evidence. Domain B (Wild-Sim) applies the same operators to 700 in-the-wild colour face crops [37] of low and varying resolution with an unknown JPEG history. It is never used for training and serves only as the target of the cross-domain protocol.

Attack taxonomy. Eight classes cover the two attack groups (Fig. 2). Each operator takes a strength $s \in (0, 1]$ that scales the size of every artifact and returns a soft mask $M \in [0, 1]^{H\times W}$ of the pixels that carry injected evidence.

- **print** (physical) uses Yule–Nielsen tone transfer with dot gain, an ordered halftone screen, paper grain, loss of live fine texture, a small perspective shift and a re-capture chain with sensor noise and JPEG compression.
- **replay** (physical) uses display gamut and contrast compression, an RGB sub-pixel stripe pattern that turns into a moiré beat, rolling-shutter bands, backlight glare and re-capture with double compression.

- **mask3d** (physical) smooths the skin region, keeps eye and mouth holes through which real features remain visible, adds a waxy sheen and adds a seam at the mask edge.
- **faceswap** (digital) aligns a donor face to the target, passes it through a low generator resolution, upsamples it, transfers colour and blends it into the inner face. The remaining evidence is the blending edge and resolution mismatch.
- **reenact** (digital) applies a smooth warp to the mouth and eye regions with a local change in sharpness and contrast.
- **synth** (digital) adds over-smoothed skin with over-sharpened detail, a transposed-convolution checkerboard [18] and a radial peak in the Fourier magnitude [7].
- **adversarial** (digital) is a real ℓ_∞ PGD perturbation ($\varepsilon = 6/255$, 7 steps) [38] made against a separate small VGG-style surrogate trained only on the training split. The test perturbations are therefore a black-box transfer attack. The surrogate itself is flipped to the genuine class on 38.8% of the crafted samples.

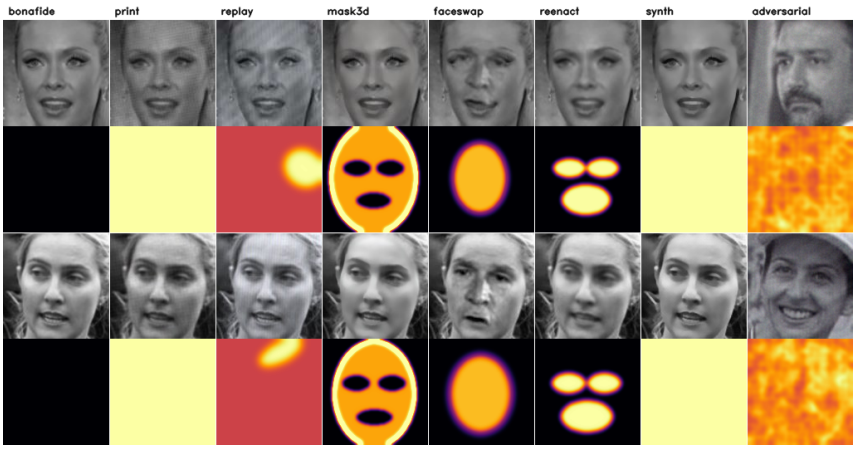


Fig. 2. Attack taxonomy and masks

For two source identities the upper strip of Fig. 2 shows the eight classes and the lower strip shows the evidence mask used to train and score the localiser. Print and replay change the whole capture. 3-D mask, face-swap, reenactment and adversarial attacks have a local support.

Difficulty calibration. A simulated benchmark is useless if a simple probe can solve it. The strength s is swept and the largest value is chosen at which a reference detector (CDCN++) does *not* saturate on the binary or the eight-way task. This gives $s = 0.62$, where the reference detector reaches 1.92% ACER but only 91.0% eight-way accuracy and 3.74% calibration error, with a worst per-family APCER of 2.67%. The binary task is close to, but not at, the ceiling, while family discrimination, calibration and unseen attacks remain open. This matches the pattern reported for UniAttackData [9].

Protocols. In P1 (seen attacks) all eight families appear in train, validation and test with identity-disjoint splits. In P2 (leave-one-attack-out) one family is removed from training and appears only at test time. P2 reports ACER, the detection rate of the held-out family and the *novelty AUROC*, which measures how well the uncertainty of the model ranks the held-out family above the attack families seen in training. In P3 (cross-domain) the model is trained on Domain A and tested on Domain B without any adaptation. All thresholds are fixed at the equal-error point of the validation split. Following ISO/IEC 30107-3 [39], APCER is the worst error over attack families, BPCER is the rate of rejected genuine faces and

$$\text{ACER} = \tfrac{1}{2}\big(\text{APCER} + \text{BPCER}\big). \tag{2}$$

Eq. (2) is the main error measure in all tables.

C. *Central-Difference and Spectral Stem*

Central Difference Convolution [2] mixes a normal convolution with its local gradient response,

$$y(p_0) \;=\; \sum_{p\in\mathcal{N}} w(p)\,x(p_0+p) \;-\; \theta\Big(\sum_{p\in\mathcal{N}} w(p)\Big)x(p_0). \tag{3}$$

The second term of Eq. (3) strengthens the part of the response that does not depend on brightness and costs only one 1×1 convolution. Four strided CDC stages reduce a 112^2 input to 7^2 and pass the 28^2 and 14^2 maps to the localisation decoder.

Periodic structure such as a display grid or an upsampling checkerboard is lost in a 7×7 token grid. A small spectral branch is therefore added. The log-magnitude DFT of the brightness channel is pooled onto a polar grid of 16 radii and 12 angles, which does not change with translation. A two-layer MLP turns this profile into four spectral tokens that are added to the

spatial tokens. The same profile is copied into the evidence record so that the peak frequency, its strength and the direction spread can be quoted in the explanation.

D. Adaptive Quantum Attention

Amplitude encoding. For a head of dimension d, queries and keys are scaled to unit length, $|\psi_i^Q\rangle = \boldsymbol{q}_i/\|\boldsymbol{q}_i\|$ and $|\psi_j^K\rangle = \boldsymbol{k}_j/\|\boldsymbol{k}_j\|$. This is a real amplitude encoding of $d = 2^n$ values over n notional qubits.

Rotation circuit. A depth-L circuit acts on a ring of amplitudes. Each layer rotates disjoint neighbouring pairs, alternating the pairs between layers, with one learnable angle per rotation,

$$G(\vartheta) = \begin{pmatrix} \cos\vartheta & -\sin\vartheta \\ \sin\vartheta & \cos\vartheta \end{pmatrix}, \qquad U(\boldsymbol{\vartheta}) = \prod_{\ell=1}^{L} \bigotimes_{k} G(\vartheta_{\ell k}). \tag{4}$$

The circuit in Eq. (4) is the real-valued form of a hardware-efficient ansatz with nearest-neighbour gates [27]. Since U is a product of rotations it stays exactly orthogonal during training. The measured norm error stayed below 10^{-6}, so no extra penalty is needed. It costs only $L\lfloor d/2 \rfloor$ angles, which is $\mathcal{O}(dL)$ instead of $\mathcal{O}(d^2)$ [27], [29]. Because it keeps vector length unchanged the score below is a true probability.

Born channel. Queries and keys are rotated by two *different* circuits, since a shared circuit would cancel out of the inner product. The score is the Born-rule transition probability,

$$S_{ij} \;=\; \left|\langle \psi_i^Q | U_Q^\top U_K | \psi_j^K \rangle\right|^2 \;\in\; [0,1]. \tag{5}$$

The score in Eq. (5) is quadratic. It does not change when either state flips sign and it is equally high for alignment or opposite alignment. An interference pattern has exactly this property, because a moiré fringe and its negative carry the same evidence, whereas a dot product treats them as opposites.

Adaptive gating. Not every kind of evidence is interference-like. A blending edge, for example, is a plain edge. A gate for each head, computed from simple statistics of the token set, decides the mix of the two channels,

$$\boldsymbol{g} = \sigma\big(\boldsymbol{W}_g\,[\,\mathrm{mean}_t(\boldsymbol{X}) \,\|\, \mathrm{std}_t(\boldsymbol{X})\,] + \boldsymbol{b}_g\big) \in [0,1]^{H_{\text{heads}}}, \tag{6}$$

$$\boldsymbol{A} = \mathrm{softmax}\Big((1-g_h)\tfrac{\boldsymbol{Q}\boldsymbol{K}^\top}{\sqrt{d}} \;+\; g_h\,\beta\,\boldsymbol{S}\Big), \tag{7}$$

with β a learnable temperature. In Eq. (6) $\boldsymbol{W}_g$ starts at zero and $\boldsymbol{b}_g$ at -2, so the gate starts near 0.12. Every block uses LayerScale [40] with residual gains that start at 10^{-2}. The network therefore begins as a plain convolutional model and slowly adds attention, with the Born channel of Eq. (7) gaining weight only as training goes on. In tests with an equal budget, removing LayerScale let the new attention add noise to the stem and slowed training. The final gate value is reported in the explanation, which makes the mechanism open to inspection.

Entropy band. Let $\mathcal{H} = \mathbb{E}_t\big[-\sum_j A_{ij}\log A_{ij}\big]$ be the mean row entropy of the attention matrix over T tokens. A very low value means attention has collapsed onto one token. A very high value means it has spread evenly and carries no information. A two-sided hinge keeps it inside a band,

$$\mathcal{L}_{\text{coh}} = \big[\lambda_{\text{lo}}\log T - \mathcal{H}\big]_+ + \big[\mathcal{H} - \lambda_{\text{hi}}\log T\big]_+, \tag{8}$$

where Eq. (8) uses $(\lambda_{\text{lo}}, \lambda_{\text{hi}}) = (0.30, 0.80)$.

E. Quantum Evidence Basis

Let $\{|e_b\rangle\}_{b=1}^{B}$ be the first B standard basis vectors and $V(\boldsymbol{\phi})$ a second rotation circuit of the same kind as Eq. (4). The evidence basis is $|\phi_b\rangle = V(\boldsymbol{\phi})|e_b\rangle$, which stays exactly orthonormal for every $\boldsymbol{\phi}$. Measuring the output token states against it gives the Born probabilities

$$\Pi_{tb} \;=\; \left|\langle \phi_b | \psi_t \rangle\right|^2, \qquad \textstyle\sum_b \Pi_{tb} \le 1. \tag{9}$$

The tensor $\boldsymbol{\Pi}$ of Eq. (9) has two jobs. Its average over tokens is added to the classification features. The same tensor, not an approximation of it, is exported as the explanation. A small extra classifier trained on $\overline{\boldsymbol{\Pi}}$ with cross-entropy pushes each mode to respond to one class, so that after training each mode can be *named* after the family it responds to most (Table XIV). These names are read from the trained model and are not assigned by hand.

F. *Decoupled Evidential Head*

A Dirichlet distribution is fully described by a direction on the simplex and a concentration. The standard evidential method [21] predicts all K concentrations together. Here the two parts are predicted separately,

$$\boldsymbol{\pi} = \mathrm{softmax}(\boldsymbol{W}_c\, \boldsymbol{h}), \tag{10}$$

$$E = \exp\big(\mathrm{clip}(f_e(\mathrm{sg}[\boldsymbol{z}]))\big), \tag{11}$$

where Eq. (10) gives the class direction, Eq. (11) gives the amount of evidence and $\mathrm{sg}[\cdot]$ is a stop-gradient. Following subjective logic [22], the concentrations, total strength and vacuity are

$$\boldsymbol{\alpha} = \mathbf{1} + E\,\mathrm{sg}[\boldsymbol{\pi}], \qquad S = K + E, \qquad u = \frac{K}{S}. \tag{12}$$

From Eq. (12) the belief masses are $b_k = E\pi_k/S$ and the projected probability under a uniform base rate is $\hat{p}_k = (1 + E\pi_k)/(K + E)$. The direction $\boldsymbol{\pi}$ is reported as the belief and u as the uncertainty.

This split has three effects that were checked in tests with equal budgets. First, $\arg\max_k \hat{p}_k = \arg\max_k \pi_k$, so the class decision is trained with plain cross-entropy. The joint form sends every gradient through $\boldsymbol{\alpha}$, where the usual KL term pushes the pre-activation negative until the gradient vanishes. This *evidence starvation* was measured directly. Replacing the joint head with the split head recovered several points of eight-way accuracy. Second, adding uncertainty cannot reduce detection accuracy, because the magnitude branch only reads a stop-gradient copy of the shared features. Third, suppressing evidence becomes a one-dimensional problem. All concentration lives in the single number E, so an out-of-taxonomy constraint cannot bend the class direction.

The evidential loss joins cross-entropy on the direction, a Brier fit on the concentration [21] and a term that ties the evidence to correctness,

$$\mathcal{L}_{\mathrm{ev}} = \mathrm{CE}(\boldsymbol{\pi}, y) \;+\; \tfrac{1}{2}\,\mathbb{E}\big[\|\boldsymbol{y} - \hat{\boldsymbol{p}}\|^2\big] \;+\; \lambda_c\,\mathrm{BCE}\big(1 - u,\, \mathbb{1}[\hat{y} = y]\big), \tag{13}$$

with $\lambda_c = 0.4$. The Brier form in Eq. (13) is used instead of the digamma-based likelihood because the gradient of the latter almost vanishes while the Dirichlet is still close to uniform, which is the state of a newly started model. The usual KL term is dropped. In the split head the wrong-class evidence is already $E\pi_k$, which cross-entropy drives to zero by sharpening $\boldsymbol{\pi}$. Keeping the KL term made short-schedule models saturate at $u \approx 1$ everywhere. The correctness term trains $1 - u$ directly as the probability that the decision is right and makes vacuity useful within about two epochs. In a leave-one-attack-out test this term moved the AUROC with which vacuity separates a held-out family from genuine samples from 25.6% (inverted) to 73.9%.

G. *Pseudo-Unseen Evidence Suppression*

A trained stand-in for unknown inputs gives better uncertainty than an internal score alone [23], but attack detection has no natural pool of outliers. PUES builds one from the taxonomy itself. Two training images $\boldsymbol{x}_a, \boldsymbol{x}_b$ from *different* attack families are mixed with $\lambda \sim \mathrm{Beta}(2, 2)$ as $\tilde{\boldsymbol{x}} = \lambda \boldsymbol{x}_a + (1 - \lambda)\boldsymbol{x}_b$. No single physical or generative process can produce such a blend, so it lies outside the taxonomy. Its evidence is kept well below that of real samples,

$$\mathcal{L}_{\mathrm{PUES}} = \frac{\big[\,\overline{\|\tilde{\boldsymbol{e}}\|_1} \;-\; \rho\,\overline{\|\boldsymbol{e}\|_1}^{\,\mathrm{sg}}\,\big]_+}{\overline{\|\boldsymbol{e}\|_1}^{\,\mathrm{sg}} + 1}, \qquad \rho = 0.25, \tag{14}$$

where bars in Eq. (14) are batch means. The *relative* form matters. An absolute target $\|\tilde{\boldsymbol{e}}\|_1 \to 0$ is met trivially by giving zero evidence everywhere. This failure was seen in an early version, which fell to 12.5% accuracy over eight classes. Tying the target to the evidence of real samples in the same batch removes the shortcut. Unlike mixup [41], which asks for a mixed *label*, PUES asks for an *absence of belief*, which is only possible because the head is evidential.

H. *Evidence Localisation*

A light feature-pyramid decoder joins the 7^2 token map with the 14^2 and 28^2 stem maps and predicts $\boldsymbol{M}$ at half the input resolution. It is trained against the ground-truth mask $\boldsymbol{M}^\star$ of the operator with

$$\mathcal{L}_{\mathrm{loc}} = \mathrm{BCE}(\boldsymbol{M}, \boldsymbol{M}^\star) + \mathrm{Dice}(\boldsymbol{M}, \boldsymbol{M}^\star), \tag{15}$$

where the Dice term of Eq. (15) follows [42]. Unlike depth supervision [1], [3] this target is the actual support of the evidence and is defined for digital attacks as well as physical ones.

I. Risk Calibration

Attack families are not equally harmful. With severity weights $\boldsymbol{\varsigma}$, an unknown-attack severity ς_{unk} and evidence area $a = \overline{M}$,

$$r = (1 - u) \sum_k \hat{p}_k \varsigma_k \; + \; \lambda_u \, u \, \varsigma_{\text{unk}} \; + \; \lambda_a \, a. \tag{16}$$

The first term of Eq. (16) is the expected harm under the posterior, reduced when evidence is low. The second term adds a premium for unknown attacks, so an input that is clearly unknown is escalated instead of passed. The third term rewards evidence that is spread over a clear region. All coefficients and the band edges that turn r into a Low, Medium, High or Critical label are set as deployment policy and are not learned, so that an operator can inspect and adjust them. A vacuity of $u \geq 0.55$ sends the sample to a human reviewer.

J. Evidence-Grounded Reasoning

Every forward pass writes a typed evidence record holding the posterior $\hat{\boldsymbol{p}}$, the vacuity u, the risk score and band, the top facial regions of $\boldsymbol{M}$ with their mean response, the area and spread of $\boldsymbol{M}$, the three strongest quantum modes with their names and activations, the spectral descriptors and the mean gate value. Text is generated *only* from this record. The grounded generator used for all results in this paper is deterministic. Each sentence is emitted by a rule whose guard is a numeric condition on the record, so every sentence traces back to a measured value and no fact can be invented. It runs in under a millisecond with no network call. An optional back-end hands the record to a hosted multimodal language model with an instruction to paraphrase only, then checks every number in the result against the record. This reverses the usual setup of language model forensics [14], [15], in which the language model sees the image but not the state of the detector.

K. Training Objective

The full loss is

$$\mathcal{L} = \mathcal{L}_{\text{ev}} + w_m \mathcal{L}_{\text{loc}} + w_b \mathcal{L}_{\text{basis}} + w_s \mathcal{L}_{\text{sup}} + w_p \mathcal{L}_{\text{PUES}} + w_c \mathcal{L}_{\text{coh}}, \tag{17}$$

where Eq. (17) uses $(w_m, w_b, w_s, w_p, w_c) = (0.6, 0.25, 0.3, 0.15, 0.02)$ and combines Eqs. (8), (13), (14) and (15). The term $\mathcal{L}_{\text{basis}}$ is the cross-entropy of the mode classifier and $\mathcal{L}_{\text{sup}}$ is a cross-entropy over the three super-classes (genuine, physical and digital) found by summing $\hat{p}$ within each group. A mix-up between two families in the same super-class is therefore punished less than one that crosses the physical and digital boundary.

For the main seen-attack results three more terms act on the predicted distribution and add no parameters. The classification term is scaled by the focal factor $(1 - \hat{p}_y)^\gamma$ with $\gamma = 2$ [43]. A binary cross-entropy $\mathcal{L}_{\text{bin}}$ on the attack probability $1 - \hat{p}_0$ (weight 0.5) protects the attack or genuine decision on which ACER is based. A pair penalty

$$\mathcal{L}_{\text{pair}} = \sum_{(c,c')} w_{cc'} \, \mathbb{E}_{i \,|\, y_i = c} \big[\hat{p}_{c'}(x_i) \big] \tag{18}$$

reduces the confusions seen during development. In Eq. (18) face-swap with reenactment and 3-D mask with synthetic are punished in both directions with $w = 0.15$, while replay with print and face-swap with print use $w = 0.05$. Each pair is averaged over its own samples. The baselines get the same focal factor and binary term together with a pooled pair penalty over the first two pairs with weight 0.15.

IV. Results and Discussion

A. Experimental Setup

Every model is trained from scratch without pretrained weights, so the comparison is between designs and losses and not between pretraining datasets. Training budgets differ between protocols, so models are compared only within a table.

For P1 (Table III) X-UAD, CDCN++ and ResNet-18 are trained for 12 epochs on the full 19,200-sample training split with the focal factor, binary term and pair penalty. The kept checkpoint is the epoch with the best validation eight-way accuracy plus macro-F1 among the epochs whose validation ACER is within 0.5 points of the best so far. Seven X-UAD variants were trained under this budget and the one with the lowest validation ACER is reported. Each baseline was trained once, so X-UAD received more tuning. ViT-Tiny was not retrained under this recipe. At the original 10-epoch budget it stays near chance and its row in Table III comes from that run.

The other protocols use the original recipe without the three extra terms. P3 reuses the 10-epoch P1 checkpoints. P2 and its open-set ablation use 5 epochs on a fixed $4{,}800$-sample subset ($4{,}382$ after the held-out family is removed) and the component ablation uses 5 epochs on $3{,}592$ samples. All runs use batch size 48, AdamW (lr $= 2.5 \times 10^{-3}$, weight decay 0.02) with a one-cycle schedule, gradient clipping at 3.0 and 112×112 random crops from 128 px images with flips and mild brightness changes. Geometric warps are not used because they would destroy the periodic patterns. Everything runs on CPU with a single seed (0). Models are compared with McNemar's exact test on paired errors, which says nothing about variation across seeds.

The first baseline is ResNet-18 [44], a residual CNN of matched width. The second is ViT-Tiny [45], a patch-16 transformer with normal attention as in the ViT-B/16 entry of [9]. The third is CDCN++ [2], [3], a central-difference network with pixel-wise map supervision. CDCN++ shares the stem of X-UAD, so comparing against it shows what the transformer, the Born channel, the evidential head and the reasoning module add.

B. Seen Attacks (P1)

TABLE III
RESULTS ON P1

Method	Params (M)	ACER↓ (%)	APCER↓ (%)	BPCER↓ (%)	AUC↑ (%)	EER↓ (%)	ACC_8 ↑ (%)	$F1_{macro}$ ↑ (%)	ECE↓ (%)	ECE_{TS} ↓ (%)
ResNet-18 [44]	1.59	1.88	3.08	0.67	99.95	0.67	**96.93**	**96.92**	**2.35**	**0.34**
ViT-Tiny [45]†	0.64	49.75	60.67	38.83	73.43	34.65	51.27	46.04	0.67	–
CDCN++ [2]	0.53	0.46	0.42	**0.50**	**99.99**	**0.33**	95.19	95.17	5.07	0.46
X-UAD (proposed)	1.29	**0.42**	**0.33**	**0.50**	99.98	0.42	94.06	94.06	5.17	0.60

Table III uses 9,600 identity-disjoint test samples. ACC_8 is eight-way family accuracy and ECE_{TS} is ECE after temperature scaling fitted on the validation split. Bold marks the best of the three detectors trained under the common recipe. ViT-Tiny (marked †) uses the original 10-epoch recipe and its low ECE reflects a detector near chance.

P1 is close to saturated for every model with a convolutional front end. ResNet-18, CDCN++ and X-UAD all reach at least 99.95% AUC and at most 1.9% ACER, which matches the seen-attack results of [9]. On the attack or genuine decision X-UAD has the lowest ACER at 0.42% against 0.46% for CDCN++. The two models disagree on only 22 test samples, 10 in favour of X-UAD and 12 against ($p = 0.83$), so this is a tie. Against ResNet-18 the difference is clear with 74 discordant samples in favour of X-UAD against 12 ($p = 4.7 \times 10^{-12}$). Table IV lists the APCER of each family, where the ISO APCER is the worst column. ResNet-18 misses 3.1% of face-swaps, while the worst family of X-UAD is replay at 0.33%.

TABLE IV
PER-FAMILY APCER (%)

Method	Print	Replay	3D mask	Face-swap	Reenact	Synthetic	Adversarial	Worst
ResNet-18 [44]	0.33	0.33	0.00	3.08	1.33	0.75	0.00	3.08
ViT-Tiny [45]†	40.25	0.00	4.42	60.67	60.00	59.67	0.00	60.67
CDCN++ [2]	0.08	0.00	0.00	0.42	0.17	0.00	0.00	0.42
X-UAD (proposed)	0.08	0.33	0.00	0.17	0.17	0.08	0.00	0.33

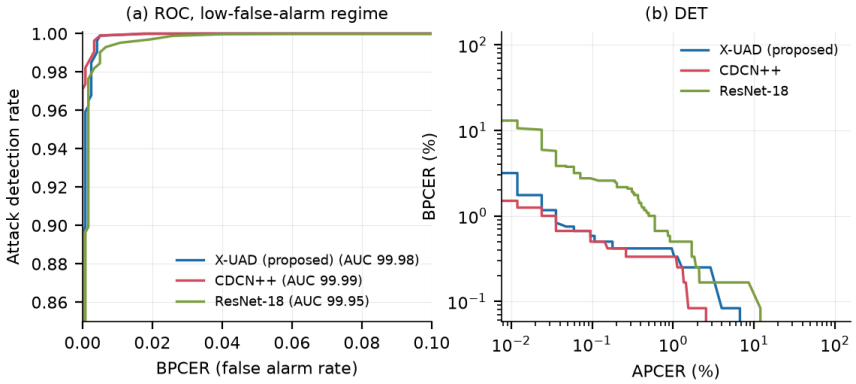


Fig. 3. ROC and DET curves

The eight-way decision goes the other way. ResNet-18 reaches 96.93% and CDCN++ 95.19% against 94.06% for X-UAD. Both differences are significant (443 against 168 discordant samples with $p = 1.6 \times 10^{-29}$ and 349 against 241 with $p = 1.0 \times 10^{-5}$). Fig. 4 gives the row-normalised confusion matrix of X-UAD. The largest errors are synthetic predicted as 3-D mask (7.5%), face-swap and reenactment confused both ways (5.6% and 5.4%) and 3-D mask predicted as synthetic (4.6%), which are the pairs targeted by Eq. (18). Two of the larger errors, synthetic as 3-D mask and face-swap as print (4.0%), cross the physical and digital boundary. Fig. 3 shows the ROC curve in the low false alarm range together with the DET curve.

X-UAD shows no calibration advantage for its class posterior. Before correction ResNet-18 has the lowest ECE (2.35% against 5.07% for CDCN++ and 5.17% for X-UAD). After temperature scaling [46] the three reach 0.34%, 0.46% and 0.60%

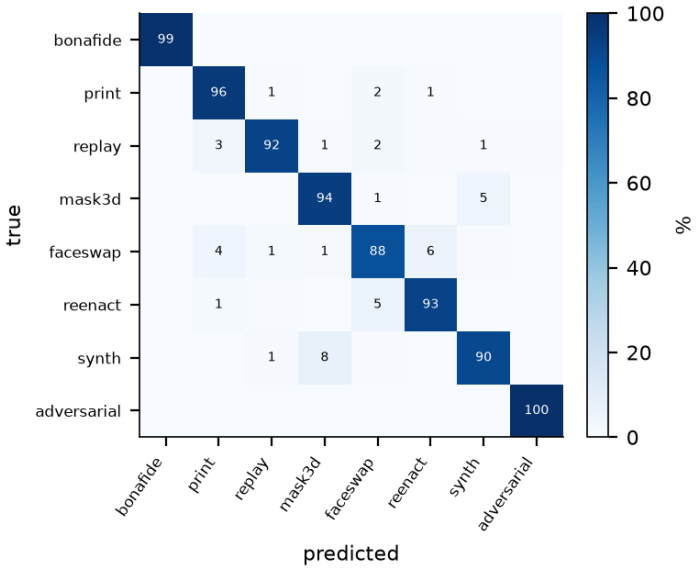


Fig. 4. Confusion matrix of X-UAD

in the same order. What X-UAD adds is a different quantity, the vacuity of Eq. (12), which is studied as an open-set score in P2. The plain patch-16 transformer fails at 49.8% ACER. With 16×16 patches it never sees the fine texture of the physical families, which is why X-UAD places a central-difference stem under its transformer.

C. *Unseen Attack Families (P2)*

P2 tests what happens when an attack family never seen in training appears at test time. Table V holds out one family at a time and reports ACER, eight-way accuracy, the detection rate of the held-out family and the novelty AUROC. Means are taken over all four folds and novelty AUROC values below 50 mean an inverted ranking.

TABLE V
RESULTS ON P2

Metric	Method	Replay	3D mask	Face-swap	Adversarial	Mean
ACER (%)↓	ResNet-18 [44]	51.96	30.38	49.96	30.71	40.75
	CDCN++ [2]	25.67	39.92	25.25	25.58	29.10
	X-UAD (proposed)	26.79	21.42	38.33	39.67	31.55
ACC_8 (%)↑	ResNet-18 [44]	53.94	62.66	51.39	61.53	57.38
	CDCN++ [2]	60.62	65.07	73.23	63.84	65.69
	X-UAD (proposed)	60.95	69.01	73.34	57.59	65.22
Unseen detection rate (%)↑	ResNet-18 [44]	100.00	61.67	37.17	99.92	74.69
	CDCN++ [2]	93.08	34.25	60.33	93.08	70.19
	X-UAD (proposed)	88.00	88.08	35.58	98.17	77.46
Novelty AUROC (%)↑	ResNet-18 [44]	35.29	56.69	74.40	36.88	50.82
	CDCN++ [2]	49.28	66.57	81.66	55.08	63.15
	X-UAD (proposed)	71.03	50.41	87.86	43.04	63.09

Mean ACER lies between 29 and 41% for every method, against under 2% in P1. This should *not* be read only as the cost of holding out a family, since P2 uses a quarter of the data for half the epochs. The ablation shows the full X-UAD already at 38% ACER with a similar budget and every family seen (Table X). Only comparisons within Table V are valid. Within it X-UAD rejects the unseen family most often on average (77.5% against 74.7% and 70.2%, Fig. 5). Its novelty AUROC clearly leads on replay (71.0 against 49.3 and 35.3) and face-swap (87.9 against 81.7 and 74.4), trails on the other two folds and is level with CDCN++ on the mean (63.09 against 63.15).

Panel (a) of Fig. 5 shows the ACER of each fold on the full test set and panel (b) shows the detection rate of the unseen family at the validation threshold. Rejection and uncertainty ranking are reported separately because they serve different needs. Rejection matters for operation, while ranking flags that a new attack type has appeared. The adversarial fold shows the difference. X-UAD rejects 98.2% of this unseen family while still giving it low vacuity, because the perturbation sits on top of evidence the model already knows. This fold is also where the uncertainty is weakest at 43.0 novelty AUROC.

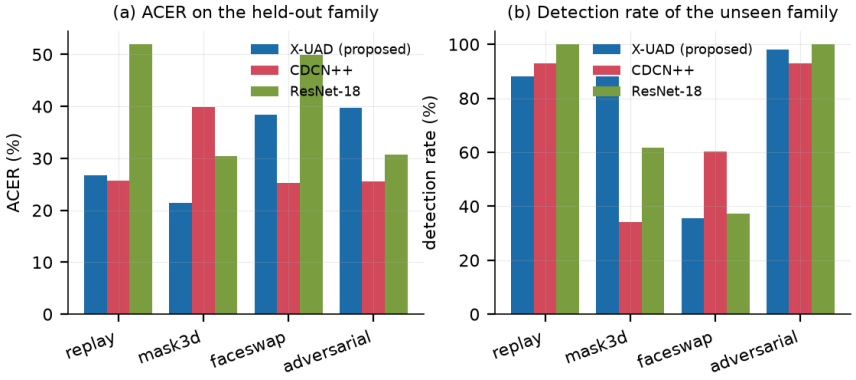


Fig. 5. Results on P2 per fold

Scoring the same X-UAD models with their own maximum softmax confidence instead of vacuity gives 55.9 mean novelty AUROC against 63.1 for vacuity. Most of this gain comes from the replay fold (71.0 against 39.5). It is small on face-swap (+1.2) and adversarial (+2.3) and negative on 3-D mask (−6.4). On 3-D mask the held-out family gets the same vacuity as the seen attacks (0.31 against 0.32). On adversarial it gets *less* (0.39 against 0.43), which explains the inversion.

D. *Cross-Domain Transfer (P3)*

Domain B differs from Domain A in colour, alignment, resolution and compression all at once and every method collapses (Table VI). With the source threshold, BPCER reaches 89 to 100%, so almost every genuine face from the new domain is rejected. By AUC, CDCN++ keeps the most signal at 65.96, while ResNet-18 (46.78), ViT-Tiny (50.18) and X-UAD (50.90) sit at or below chance. The adversarial class is excluded because it was crafted from Domain A images. This protocol is harsher than usual cross-dataset tests because four factors change together. None of the proposed mechanisms targets capture-domain shift, which would need domain generalisation methods [4], [16], [17].

TABLE VI
RESULTS ON P3

Method	ACER↓	AUC↑	EER↓	ACC_S ↑	ECE↓
ResNet-18 [44]	49.79	46.78	51.87	2.29	83.26
ViT-Tiny [45]	50.00	50.18	50.23	10.31	82.47
CDCN++ [2]	45.79	65.96	37.93	18.67	60.09
X-UAD (proposed)	49.71	50.90	49.98	3.65	81.61

E. *Calibration and Localisation*

Fig. 6(a) shows the reliability diagrams before temperature scaling. All three detectors are somewhat over- or under-confident and temperature scaling removes most of it for each, so calibration does not separate the models. Fig. 6(b) shows the uncertainty each model can offer. For the softmax baselines this is $1 - \max_k p_k$. For X-UAD it is the vacuity $u = K/S$, which is trained as an output in its own right and ranks unseen families above the seen ones on two of the four folds of Table V.

Table VII gives the IoU against the mask of each operator with maps cut at 0.5. ResNet-18 and ViT-Tiny give no map. Families whose evidence covers part of the frame (3-D mask, face-swap and reenactment) are separated from those that change the whole capture (print, replay and synthetic), since the latter have nearly constant masks and a trivially high IoU. The region-level mean is the number that shows whether the localiser works. On all 9,600 test samples X-UAD reaches 80.1% region-level IoU against 73.2% for CDCN++. It is ahead on each of the three region-level families and on pixel AUC (97.8% against 96.2%, over 40 samples per family). Sweeping the cut from 0.10 to 0.90 on validation gives a best cut of 0.55 for both models, at which the test IoU moves by less than 0.4 points (80.1% and 73.5%).

TABLE VII
LOCALISATION IOU (%)

Method	Print	Replay	3D mask	Face-swap	Reenact	Synthetic	Region-level mean	Pixel AUC
CDCN++ [2]	96.4	93.9	76.5	75.4	67.7	97.7	73.2	96.2
X-UAD (proposed)	98.1	96.9	84.2	78.8	77.2	98.2	80.1	97.8

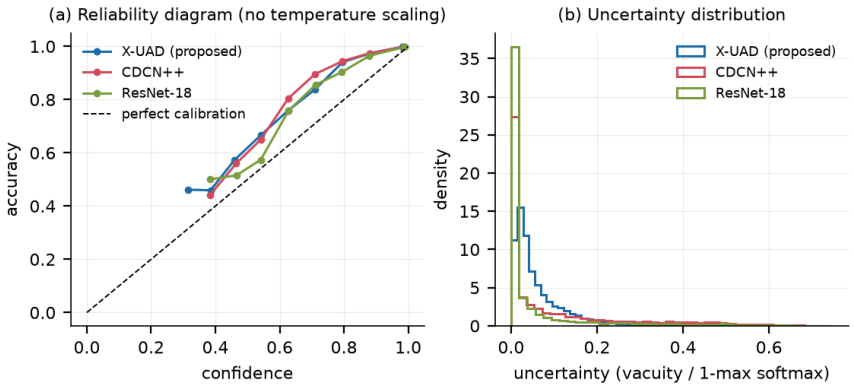


Fig. 6. Calibration and uncertainty

F. *Faithfulness of the Explanation*

An explanation is only useful if changing what it highlights changes the decision. Table VIII gives deletion and insertion AUC [26] of the attack score for the localisation map against a smoothed random field of the same scale, using 48 test samples per attack family and 95% bootstrap intervals. Insertion gain is the map score minus the control score, so an interval below zero means the map does worse than random.

TABLE VIII
FAITHFULNESS OF THE MAP

Families	n	Deletion AUC↓ map	Deletion AUC↓ random	Insertion AUC↑ map	Insertion AUC↑ random	Insertion gain, 95% CI
Region-level (3D mask, face-swap, reenact)	144	0.892	0.941	0.935	0.942	[-0.011, -0.003]
Whole-frame (print, replay, synthetic, adversarial)	192	0.904	0.943	0.921	0.942	[-0.026, -0.016]

Deletion is clean. Removing the highlighted pixels lowers the attack score faster than removing random regions on region-level families (0.892 against 0.941) and on whole-frame families (0.904 against 0.943). Insertion is not clean. Putting the highlighted pixels back into a blurred image recovers the score more slowly than random pixels in both groups. The shortfall is three times larger on whole-frame families (-0.021) than on region-level ones (-0.007) but does not vanish. The maps therefore pass the deletion test and fail the insertion test.

G. *White-Box Attacks*

The adversarial family of the benchmark is a transfer attack. Here the attacker has full access to the detector. An ℓ_∞ PGD attack [38] (10 steps, step size $\varepsilon/4$, random start) is run on 168 test attacks, 24 per family. The first goal is evasion, which pushes an attack towards the genuine class. The second goal targets X-UAD only and lowers u while a cross-entropy term holds the predicted class fixed. In Table IX the upper rows give the share of attacks still detected and the lower rows give the mean vacuity under the second attack and how often the predicted class stays the same.

TABLE IX
WHITE-BOX PGD RESULTS

ε ($\times 1/255$)	clean	1	2	4	8
CDCN++, attacks detected (%)	100.0	5.4	14.9	41.1	39.3
X-UAD, attacks detected (%)	100.0	30.4	11.3	0.0	0.0
X-UAD, mean vacuity under false-confidence attack	0.064	0.017	0.075	0.077	0.057
X-UAD, predicted class unchanged (%)	100.0	99.4	53.6	32.1	37.5

Neither detector survives evasion. At $\varepsilon = 1/255$ X-UAD still catches 30.4% of the attacks against 5.4% for CDCN++. At $2/255$ both are near 10 to 15% and from $4/255$ X-UAD detects none. CDCN++ levels off near 40%, which more likely reflects gradients that are harder to follow than true robustness. None of the evaded attacks reaches the $u \geq 0.55$ referral threshold. The second attack shows that the uncertainty can be moved on its own. At $\varepsilon = 1/255$ it cuts mean vacuity from 0.064 to 0.017 while 99.4% of predicted classes stay the same. At larger budgets the class is no longer held fixed, so the clean split is a small-perturbation effect.

H. Ablation Study

Table X removes one part at a time at a reduced budget. Deltas are relative to the full model in the first row. These rows are comparable with each other but not with Table III.

TABLE X
COMPONENT ABLATION

Variant	ACER↓	AUC↑	ACC_8 ↑	$F1_{macro}$ ↑	ECE↓	IoU_{loc} ↑
X-UAD (full, reduced budget)	38.00	87.57	69.62	69.65	2.66	64.62
w/o Adaptive Quantum Attention	32.75 (-5.25)	91.05 (+3.49)	67.45 (-2.18)	65.04 (-4.61)	3.62 (+0.96)	63.36 (-1.26)
w/o evidential head (softmax)	30.54 (-7.46)	91.06 (+3.49)	72.07 (+2.45)	71.88 (+2.23)	3.27 (+0.61)	65.02 (+0.41)
w/o spectral branch	31.13 (-6.87)	88.44 (+0.88)	70.67 (+1.04)	68.92 (-0.73)	3.12 (+0.47)	64.54 (-0.07)
w/o central-difference stem	49.08 (+11.08)	73.64 (-13.93)	48.62 (-21.00)	42.16 (-27.49)	3.03 (+0.37)	52.63 (-11.99)
w/o pseudo-unseen suppression	26.04 (-11.96)	91.08 (+3.51)	72.90 (+3.27)	72.85 (+3.20)	2.83 (+0.17)	64.23 (-0.39)
w/o localisation supervision	33.58 (-4.42)	89.17 (+1.61)	70.56 (+0.94)	68.78 (-0.87)	2.41 (-0.25)	34.34 (-30.27)

The central-difference stem matters most. Removing it costs more than all other parts together, dropping eight-way accuracy by 21 points and region-level IoU by 12. Localisation supervision gives the map its meaning. Without it region-level IoU falls by more than half while detection barely changes, so a saliency map taken for free from a detector is not the same as a supervised evidence map.

Two parts trade closed-set accuracy for uncertainty. Removing PUES *improves* reduced-budget ACER but removes the ability the term was built for. Table XI isolates this trade on the face-swap fold, where PUES is worth 5.4 novelty AUROC points and 5.2 points of family accuracy at a cost of 2.2 ACER points. AQA shows the same pattern. At reduced budget it costs ACER but improves calibration and localisation. On the face-swap fold it adds 1.9 AUC and 1.9 accuracy at equal novelty AUROC.

TABLE XI
OPEN-SET ABLATION

Variant	ACER↓	AUC↑	ACC_8 ↑	ECE↓	Unseen detect.↑	Novelty AUROC↑
Full X-UAD	38.33	91.41	73.34	6.26	35.58	87.86
w/o pseudo-unseen suppression	36.17	86.33	68.12	6.78	45.50	82.43
w/o Adaptive Quantum Attention	39.83	89.47	71.40	7.36	32.75	87.89

I. Efficiency and Gate Analysis

X-UAD adds its extra abilities at a small parameter cost (Table XII, measured on CPU for one 112×112 crop). Each rotation circuit costs $L\lfloor d/2 \rfloor$ angles against d^2 for a dense projection and the reasoning module runs in under a millisecond with no learned parameters. At 1.29M parameters X-UAD sits between ViT-Tiny and ResNet-18 in size while being the only model that gives a localisation map, named evidence modes and a written explanation. GMAC counts leave out attention products, so the X-UAD figure is slightly low.

TABLE XII
INFERENCE COST

Method	Params (M)	GMACs	Latency (ms)	Explanation outputs
ResNet-18 [44]	1.59	0.368	16.8	none
ViT-Tiny [45]	0.64	0.031	2.9	attention only
CDCN++ [2]	0.53	0.070	13.1	depth-proxy map
X-UAD (proposed)	1.29	0.119	16.9	map + modes + text

Table XIII gives the final gate g of Eq. (7), the share of the attention score taken from the Born channel, averaged over heads, layers and test samples of each family. The gate opens widest on print (0.434) and adversarial (0.394) and least on replay (0.231). Replay was expected to be the clearest interference family because of its moiré beat, so the channel is used (mean near 0.31) but the reason why it helps is not yet settled.

TABLE XIII
BORN-CHANNEL GATE PER FAMILY

Attack family	Bonafide	Print	Replay	3D mask	Face-swap	Reenact	Synthetic	Adversarial
Mean Born-channel gate g	0.309	0.434	0.231	0.287	0.267	0.267	0.284	0.394

J. Qualitative Results

The rows of Fig. 7 show, for one sample per family, the input, the predicted evidence map, the ground-truth mask and the AQA attention rollout. The localiser recovers the eye and mouth support of the reenactment and the holes of the 3-D mask and spreads over the whole frame for print, synthetic and adversarial. It is not perfect. On this face-swap it predicts an eye and mouth pattern like reenactment instead of the blending ellipse and on replay it misses the local glare spot. Table XIV lists the quantum evidence modes. Each mode is named after the family for which its Born activation of Eq. (9) is most selective (z-scored across families). These analyses and the gate values use the 10-epoch P1 model.

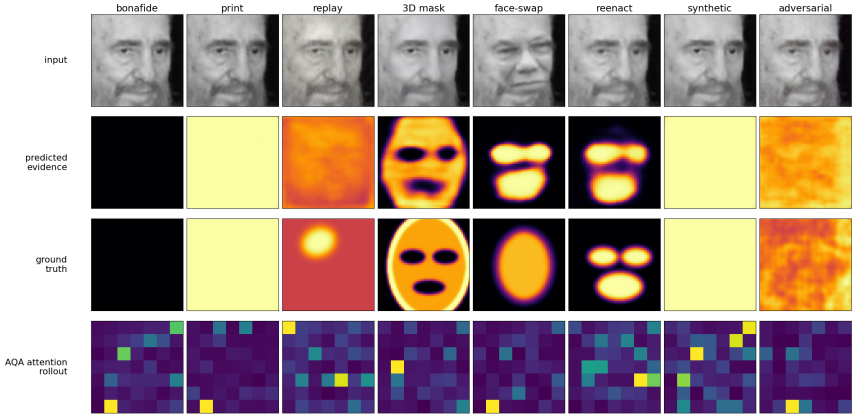


Fig. 7. Qualitative results

TABLE XIV
LEARNED EVIDENCE MODES

Mode	Most selective for	Selectivity (z)	Mean activation	Spatial dispersion
1	Replay	2.59	0.069	0.793
7	3D mask	2.51	0.051	0.810
3	Bonafide	2.45	0.058	0.835
6	Adversarial	2.36	0.057	0.869
4	Print	2.28	0.049	0.803
5	Synthetic	2.01	0.043	0.767
0	Reenact	1.85	0.047	0.783
2	Replay	1.84	0.039	0.845

V. CONCLUSION

This paper presented X-UAD, a single model that detects physical and digital face attacks, reports how uncertain it is, marks where the evidence lies and explains its decision in text that can be checked sentence by sentence. A Born-rule attention channel matches the interference nature of attack evidence at a cost of $\mathcal{O}(dL)$ parameters. An orthonormal quantum evidence basis serves both as a classification feature and as the explanation. Pseudo-unseen evidence suppression builds an outlier stand-in from the attack taxonomy and turns Dirichlet vacuity into a usable open-set score. On seen attacks X-UAD matches the strongest baseline on the attack or genuine decision and localises evidence better than every compared model. On unseen attack families it gives the highest average detection rate.

The work also has limitations. All numbers come from a single seed on a simulated benchmark and the adapters for seven public datasets have not yet been run on real captured data. X-UAD received more tuning than the baselines and the P2, P3 and ablation tables come from an earlier code version. Eight-way accuracy is lower than both convolutional baselines and there is no calibration advantage after temperature scaling. The uncertainty ranking is mildly inverted on the adversarial fold, the localisation map fails the insertion test, a white-box attacker can switch off the uncertainty and no method performs acceptably across capture domains. In future work these limits will be addressed by repeating all experiments over several seeds, running the released adapters on real captured datasets and giving the baselines the same tuning budget. Domain generalisation methods will be added to handle camera shift together with unseen attacks. Adversarial training on the vacuity output and input consistency checks will be explored to protect the uncertainty. Higher-order PUES blends will also be tested to improve the outlier stand-in. The referral threshold will be set from measured referral rates for each group.

Author Contributions

Amrutha Babu, Swethaa Premnath and Vividha Senthil Mohan contributed equally to the conceptualisation, methodology, software, construction of the benchmark, experiments, formal analysis, visualisation and writing of the original draft. Muthunagai S U guided the project administration and the editing of the manuscript and supervised the experimental process. Nivedha K provided critical review and technical validation. All authors read and approved the final manuscript.

Acknowledgment

The authors would like to express their gratitude to their supervisor, Prof. Muthunagai S U, for her invaluable mentorship, insightful guidance and support throughout the research. The authors would also like to extend their thanks to the School of Computer Science and Engineering (SCOPE) at Vellore Institute of Technology (VIT), Vellore, for its support and for providing the academic resources and facilities used in this work. This work would not have been possible without the foundational contributions of the researchers who developed and shared the Labeled Faces in the Wild (LFW) database [35], whose work has been instrumental for the broader research community.

Data Availability

The genuine faces of Domain A are taken from the Labeled Faces in the Wild (LFW) database [35], a public academic benchmark of 13,233 face photographs collected from the web. Of these, 12,794 identity-labelled images of 5,616 subjects were used. LFW is publicly available to the research community at http://vis-www.cs.umass.edu/lfw/ [36]. All attack samples were generated from these images by the benchmark operators of Section III, so no attack data was collected from people. Domain B (Wild-Sim) consists of 700 in-the-wild colour face crops [37] and is used only for testing. No new images of people were captured, no private data were collected and no attempt was made to identify any person in the images. The Wild-Sim face set, the benchmark generator, the dataset configuration files and the code are available from the corresponding author on reasonable request.

Funding

Authors have not received any fund to carry out this research work.

References

[1] Y. Liu, A. Jourabloo, and X. Liu, "Learning deep models for face anti-spoofing: Binary or auxiliary supervision," in *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 389–398.

[2] Z. Yu, C. Zhao, Z. Wang, Y. Qin, Z. Su, X. Li, F. Zhou, and G. Zhao, "Searching central difference convolutional networks for face anti-spoofing," in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 5295–5305.

[3] Z. Yu, X. Li, P. Wang, and G. Zhao, "Revisiting pixel-wise supervision for face anti-spoofing," *IEEE Trans. Biometrics, Behavior, and Identity Science*, vol. 3, no. 3, pp. 285–295, 2021.

[4] Z. Wang, Z. Wang, Z. Yu, W. Deng, J. Li, T. Gao, and Z. Wang, "Domain generalization via shuffled style assembly for face anti-spoofing," in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 4123–4133.

[5] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "Faceforensics++: Learning to detect manipulated facial images," in *Proc. IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, 2019, pp. 1–11.

[6] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, "Celeb-df: A large-scale challenging dataset for deepfake forensics," in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 3207–3216.

[7] J. Frank, T. Eisenhofer, L. Schönherr, A. Fischer, D. Kolossa, and T. Holz, "Leveraging frequency analysis for deep fake image recognition," in *Proc. Int. Conf. on Machine Learning (ICML)*, 2020, pp. 3247–3258.

[8] S.-Y. Wang, O. Wang, R. Zhang, A. Owens, and A. A. Efros, "Cnn-generated images are surprisingly easy to spot... for now," in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 8695–8704.

[9] H. Yu, H. Fang, A. Yuan, Z. Yu, A. Liu *et al.*, "Unified physical-digital face attack detection," in *Proc. 33rd Int. Joint Conf. on Artificial Intelligence (IJCAI)*, 2024, pp. 749–757.

[10] H. Yuan, A. Liu, Z. Yu, J. Wan, S. Escalera, Z. Lei *et al.*, "Unified physical-digital attack detection challenge," in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2024.

[11] Y. Ma, A. Liu, Z. Yu, J. Wan, S. Escalera, Z. Lei *et al.*, "Uniattackdata+: Unified physical-digital attack detection challenge," in *Proc. IEEE/CVF Int. Conf. on Computer Vision Workshops (ICCVW)*, 2025.

[12] S. Chen, A. Liu, Z. Yu, J. Wan, and Z. Lei, "Uniattack: Unified physical-digital face attack detection," *International Journal of Computer Vision*, 2025.

[13] J. Zhang, Z. Chen, R. Wang *et al.*, "Towards unified defense for face forgery and spoofing attacks via dual space reconstruction learning," *International Journal of Computer Vision*, 2024.

[14] Z. Huang, B. Xia, Z. Lin, Z. Mou, W. Yang, and J. Jia, "FFAA: Multimodal large language model based explainable open-world face forgery analysis assistant," *arXiv preprint arXiv:2408.10072*, 2024.

[15] X. Guo, X. Song, Y. Zhang, X. Liu, and X. Liu, "Rethinking vision-language model in face forensics: Multi-modal interpretable forged face detector," in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[16] R. Shao, X. Lan, J. Li, and P. C. Yuen, "Multi-adversarial discriminative deep domain generalization for face presentation attack detection," in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 10 023–10 031.

[17] Y. Jia, J. Zhang, S. Shan, and X. Chen, "Single-side domain generalization for face anti-spoofing," in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 8484–8493.

[18] A. Odena, V. Dumoulin, and C. Olah, "Deconvolution and checkerboard artifacts," *Distill*, 2016.

[19] D. Hendrycks and K. Gimpel, "A baseline for detecting misclassified and out-of-distribution examples in neural networks," in *Int. Conf. on Learning Representations (ICLR)*, 2017.

[20] A. Malinin and M. Gales, "Predictive uncertainty estimation via prior networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[21] M. Sensoy, L. Kaplan, and M. Kandemir, "Evidential deep learning to quantify classification uncertainty," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018, pp. 3179–3189.

[22] A. Jøsang, *Subjective Logic: A Formalism for Reasoning Under Uncertainty*. Springer, 2016.

[23] D. Hendrycks, M. Mazeika, and T. Dietterich, "Deep anomaly detection with outlier exposure," in *Int. Conf. on Learning Representations (ICLR)*, 2019.

[24] F. Liu, A. Liu, Z. Yu *et al.*, "Open-set single-domain generalization for robust face anti-spoofing," *International Journal of Computer Vision*, 2024.

[25] S. Abnar and W. Zuidema, "Quantifying attention flow in transformers," in *Proc. 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020, pp. 4190–4197.

[26] V. Petsiuk, A. Das, and K. Saenko, "RISE: Randomized input sampling for explanation of black-box models," in *British Machine Vision Conference (BMVC)*, 2018.

[27] E. A. Cherrat, I. Kerenidis, N. Mathur, J. Landman, M. Strahm, and Y. Y. Li, "Quantum vision transformers," *Quantum*, vol. 8, p. 1265, 2024.

[28] M. Comajoan Cara, G. R. Dahale, Z. Dong, R. T. Forestano, S. Gleyzer, D. Justice, K. Kong, T. Magorsch, K. T. Matchev, K. Matcheva, and E. B. Unlu, "Quantum vision transformers for quark-gluon classification," *Axioms*, vol. 13, no. 5, p. 323, 2024.

[29] T. Evans *et al.*, "From $\mathcal{O}(n^2)$ to $\mathcal{O}(n)$ parameters: Quantum self-attention in vision transformers for biomedical image classification," *arXiv preprint arXiv:2503.07294*, 2025.

[30] C.-S. Chen and E.-J. Kuo, "Quantum adaptive self-attention for quantum transformer models," *arXiv preprint arXiv:2504.05336*, 2025.

[31] Z. Zhang, J. Yan, S. Liu, Z. Lei, D. Yi, and S. Z. Li, "A face antispoofing database with diverse attacks," in *Proc. IAPR Int. Conf. on Biometrics (ICB)*, 2012, pp. 26–31.

[32] Z. Boulkenafet, J. Komulainen, L. Li, X. Feng, and A. Hadid, "OULU-NPU: A mobile face presentation attack database with real-world variations," in *IEEE Int. Conf. on Automatic Face and Gesture Recognition (FG)*, 2017, pp. 612–618.

[33] X. Guo, Y. Liu, A. Jain, and X. Liu, "Multi-domain learning for updating face anti-spoofing models," in *European Conf. on Computer Vision (ECCV)*, 2022, pp. 230–249.

[34] Y. Zhang, Z. Yin, Y. Li, G. Yin, J. Yan, J. Shao, and Z. Liu, "CelebA-Spoof: Large-scale face anti-spoofing dataset with rich annotations," in *European Conf. on Computer Vision (ECCV)*, 2020, pp. 70–85.

[35] G. B. Huang, M. Ramesh, T. Berg, and E. Learned-Miller, "Labeled faces in the wild: A database for studying face recognition in unconstrained environments," University of Massachusetts, Amherst, Tech. Rep. 07-49, 2007.

[36] University of Massachusetts, Amherst, "Labeled faces in the wild home," http://vis-www.cs.umass.edu/lfw/, 2007, dataset source for the genuine face images used in this work.

[37] A. Babu, S. Premnath, and V. Senthil Mohan, "Wild-sim in-the-wild colour face set for cross-domain testing," Released with the X-UAD code and benchmark generator, 2026.

[38] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Int. Conf. on Learning Representations (ICLR)*, 2018.

[39] ISO/IEC JTC1 SC37, "Information technology — biometric presentation attack detection — part 3: Testing and reporting," International Organization for Standardization, Tech. Rep. ISO/IEC 30107-3:2023, 2023.

[40] H. Touvron, M. Cord, A. Sablayrolles, G. Synnaeve, and H. Jégou, "Going deeper with image transformers," in *Proc. IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, 2021, pp. 32–42.

[41] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," in *Int. Conf. on Learning Representations (ICLR)*, 2018.

[42] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. Int. Conf. on 3D Vision (3DV)*, 2016, pp. 565–571.

[43] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. on Computer Vision (ICCV)*, 2017, pp. 2980–2988.

[44] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[45] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Int. Conf. on Learning Representations (ICLR)*, 2021.

[46] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. Int. Conf. on Machine Learning (ICML)*, 2017, pp. 1321–1330.